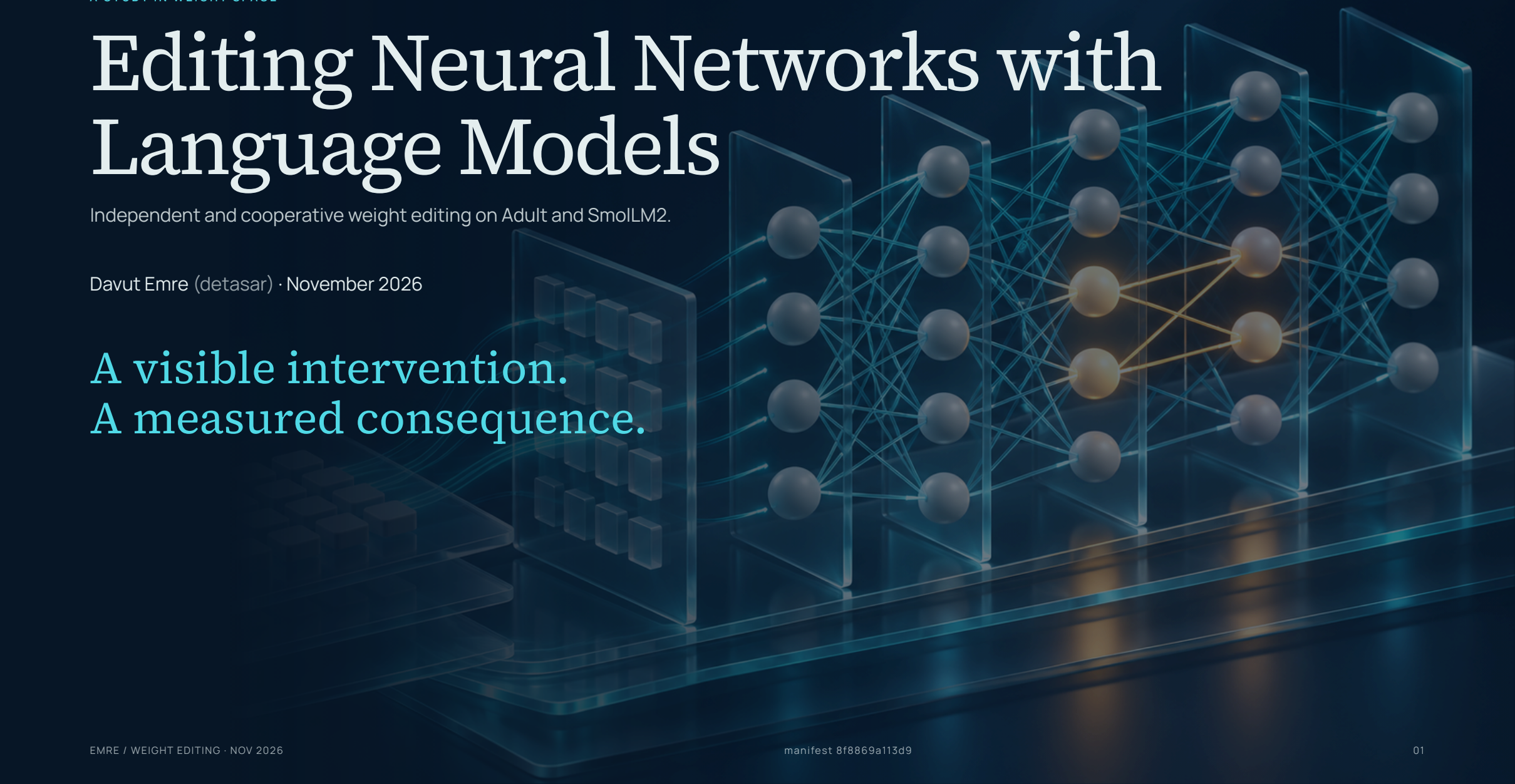
A STUDY IN WEIGHT SPACE

Editing Neural Networks with Language Models

Independent and cooperative weight editing on Adult and SmolLM2.

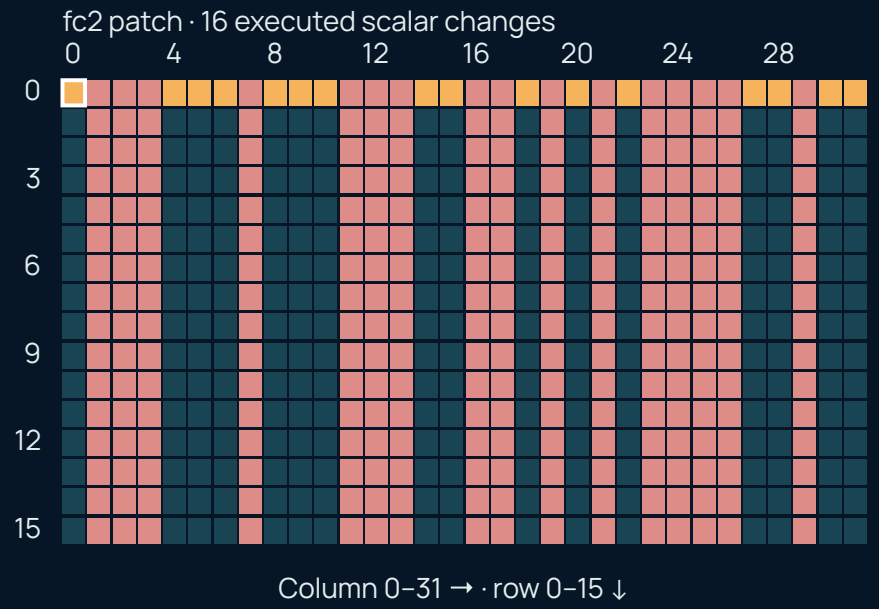
Davut Emre (detasar) · November 2026

A visible intervention.
A measured consequence.



A weight is a contribution

Changing a scalar changes a computation; improvement must still be measured.



fc2[0, 0]

Before · float32

-0.06243621558

After · float32

-0.03093621507

Amber = changed

Outline = inspected

EXECUTED ADULT PATCH · SOL

Scalar fc2[0, 0]

-0.06243621558 → -0.03093621507

16 changed · L2 0.143815856 · relative 0.048917676

$$h_1 = \mathrm{ReLU}(W_1 x + b_1)$$

x: 104 encoded inputs; W_1: 32×104 learned weights; b_1: 32 biases.

ReLU(z) = max(0,z), applied to each component. h_1: 32 hidden outputs.

Two scales, one question

A small classifier makes the intervention legible. A language model tests selection at scale.

104

Adult

Encoded predictors
Binary income classification
Permanent lesion; surviving fc2 edits

×

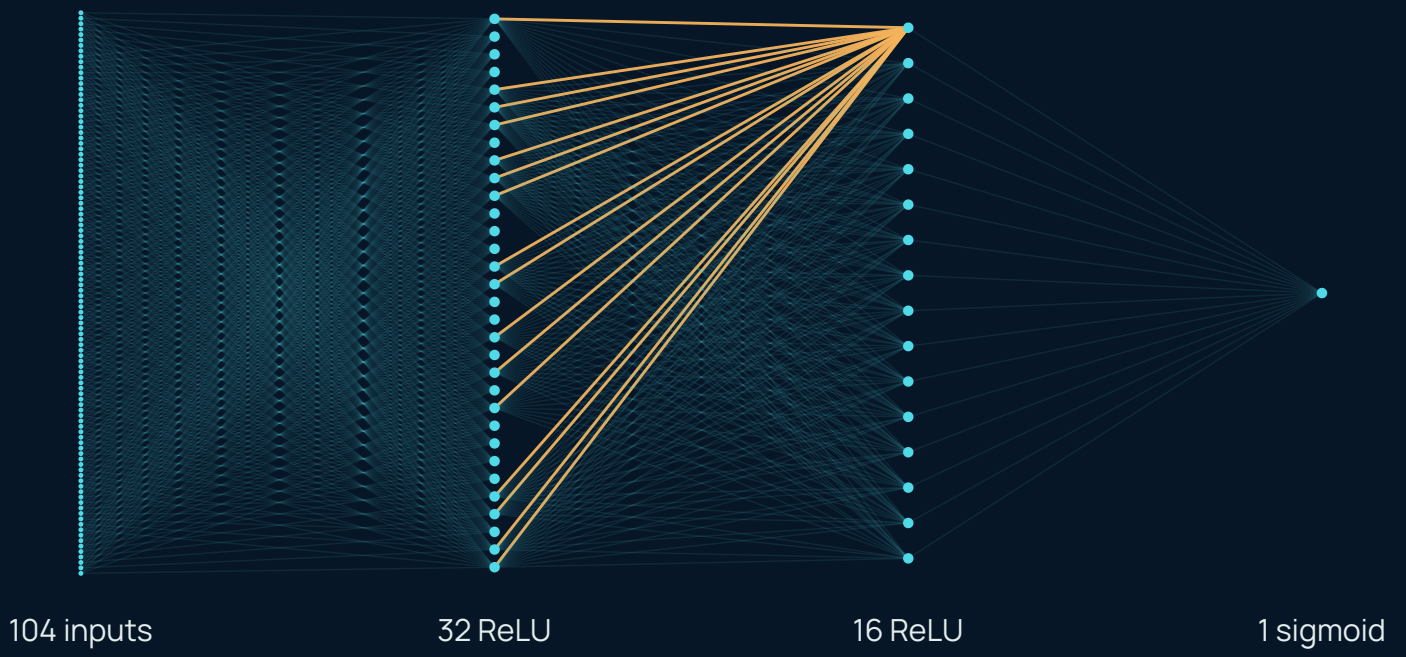
24

SmolLM2 blocks

Base language model
ARC-Easy + PIQA
Bounded down_proj edits

The whole Adult network

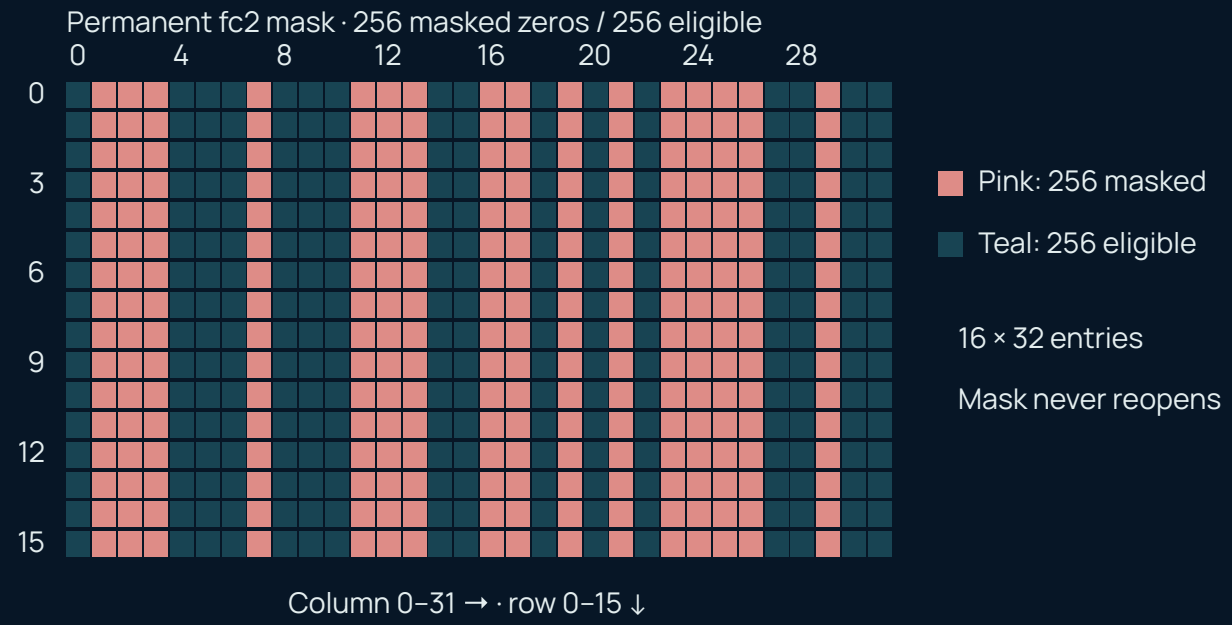
104 inputs → 32 ReLU → 16 ReLU → 1 sigmoid probability.



All 153 nodes · first-layer edges sampled · actual fc2 mask and support. Flow pulse is explanatory.

The mask stays closed

Removed connections remain zero; edits can only change surviving fc2 entries.



$$W_d = W_{clean} \odot (1 - M), \qquad \Delta \odot M = 0$$

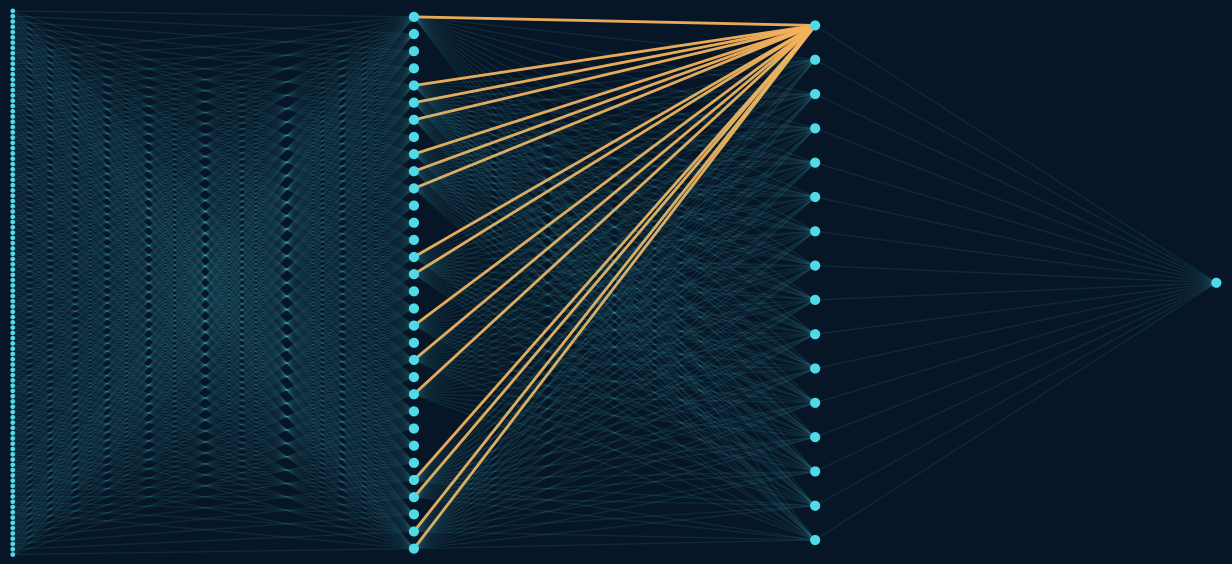
M: 16×32 binary mask; 1 means permanently removed. ⊙: elementwise product.

Wd: damaged origin from Wclean. Δ⊙M = 0 keeps every masked scalar zero.

‖Δ‖F: square root of summed squared executed changes; relative norm divides by origin norm (limit .05).

A bounded intervention

Each candidate is a complete patch relative to one immutable origin.



All 153 nodes · first-layer edges sampled · actual fc2 mask and support. Flow pulse is explanatory.

$$h_2 = \mathrm{ReLU}((W_d + \Delta)h_1 + b_2), \quad p = \sigma(W_3 h_2 + b_3)$$

h_1: 32 fixed first-layer outputs. Wd: damaged 16×32 origin; Δ: candidate change.

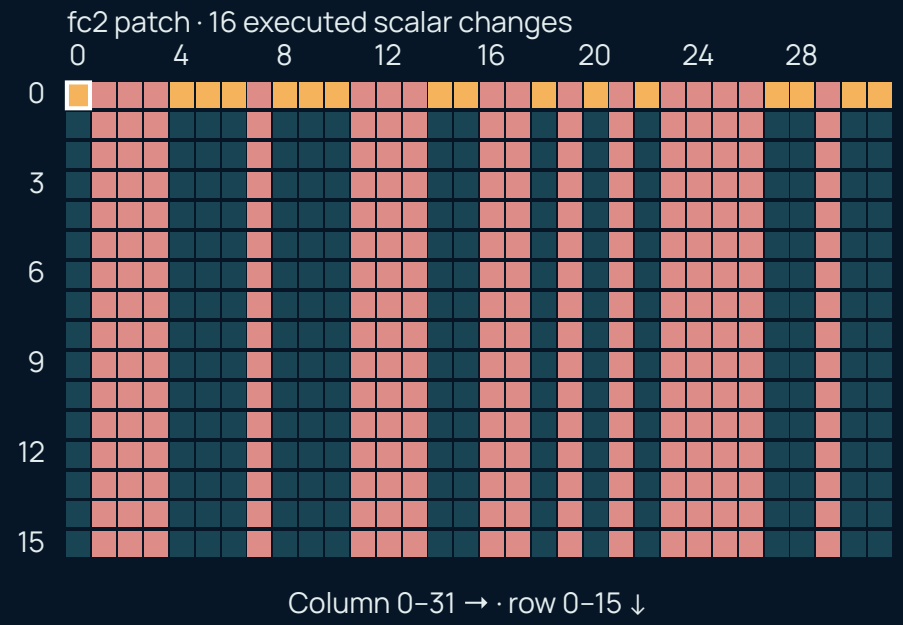
h_2: 16 ReLU outputs. W_3: 1×16 final weights; b_2 and b_3: biases.

σ(z) = 1/(1+exp(−z)); p: probability of income above 50K.

A REAL SCALAR

Before. After. Exactly.

The visual support and the scalar inspector refer to the same executed patch.



fc2[0, 0]

Before · float32

-0.06243621558

After · float32

-0.03093621507

Amber = changed

Outline = inspected

EXECUTED ADULT PATCH · SOL

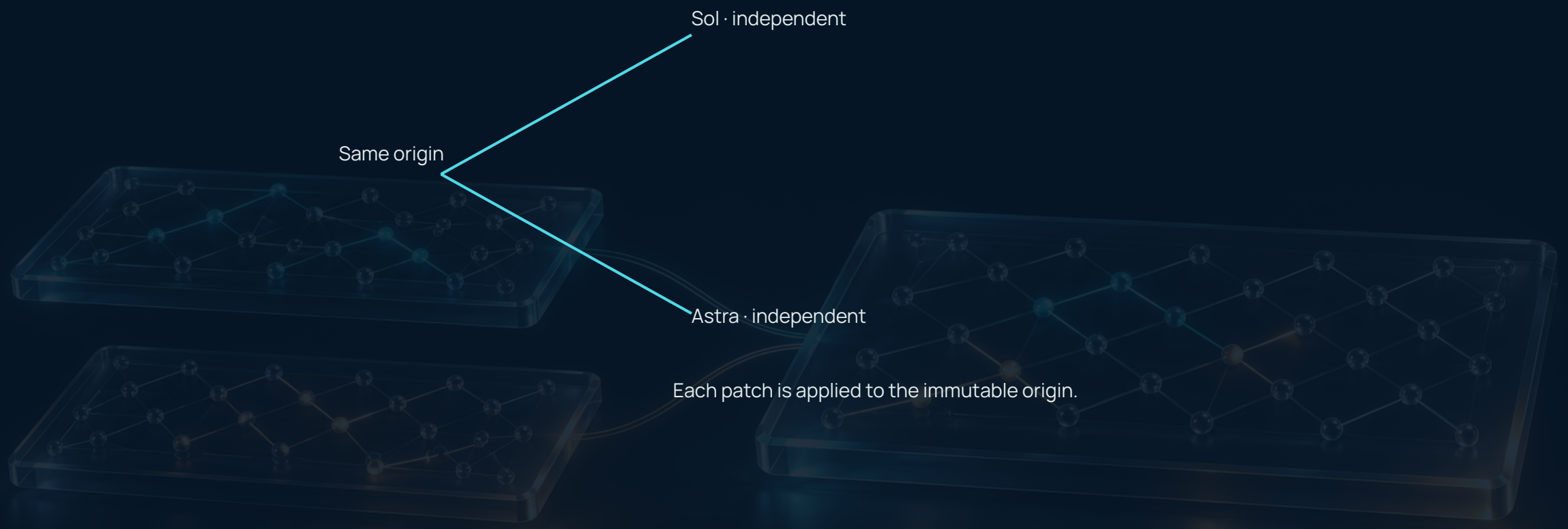
Scalar fc2[0, 0]

-0.06243621558 → -0.03093621507

16 changed · L2 0.143815856 · relative 0.048917676

Two branches from one origin

Sol and Astra never receive one another's source attempts.



Propose → validate → measure

Development feedback guides edits. Final feedback never chooses the patch.

01 **Inspect** → 02 **Propose JSON** → 03 **Validate** → 04 **DEV feedback**

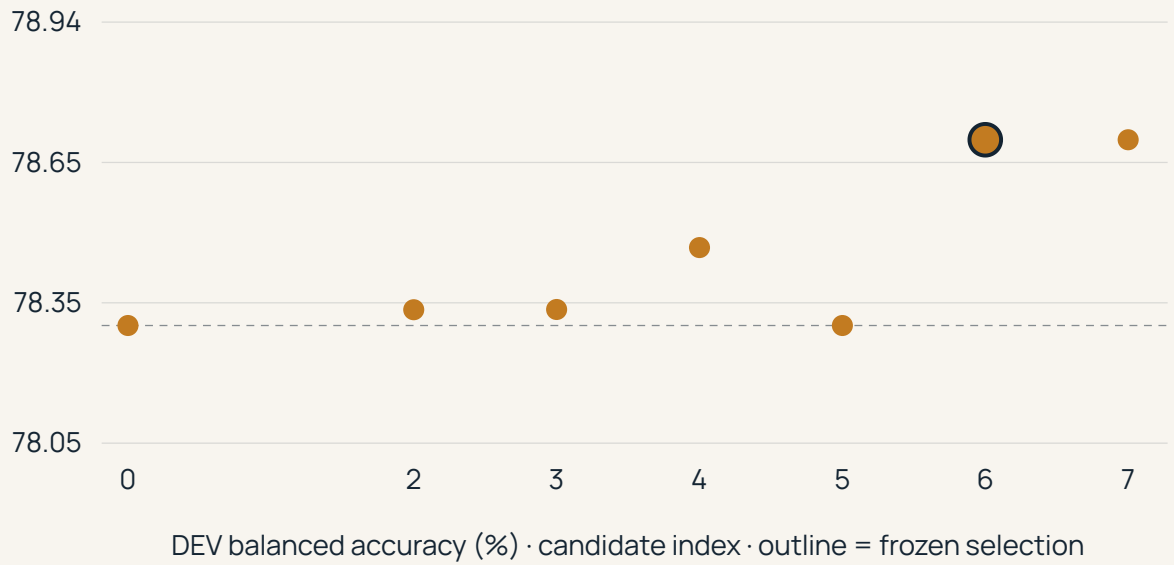
Freeze the chosen patch. Then measure FINAL.

Adult prompt · LM v2 prompt · Action schema

MEASURED DEVELOPMENT POINTS

Selection is part of the result

A higher primary development score wins; equal scores prefer the smaller edit.



Sol · candidate 6 · 16 changed scalars · L2 0.14381586 · DEV 78.694638%

$$\arg\min_i(-d_i, \|\Delta_i\|_2, \#\Delta_i, i)$$

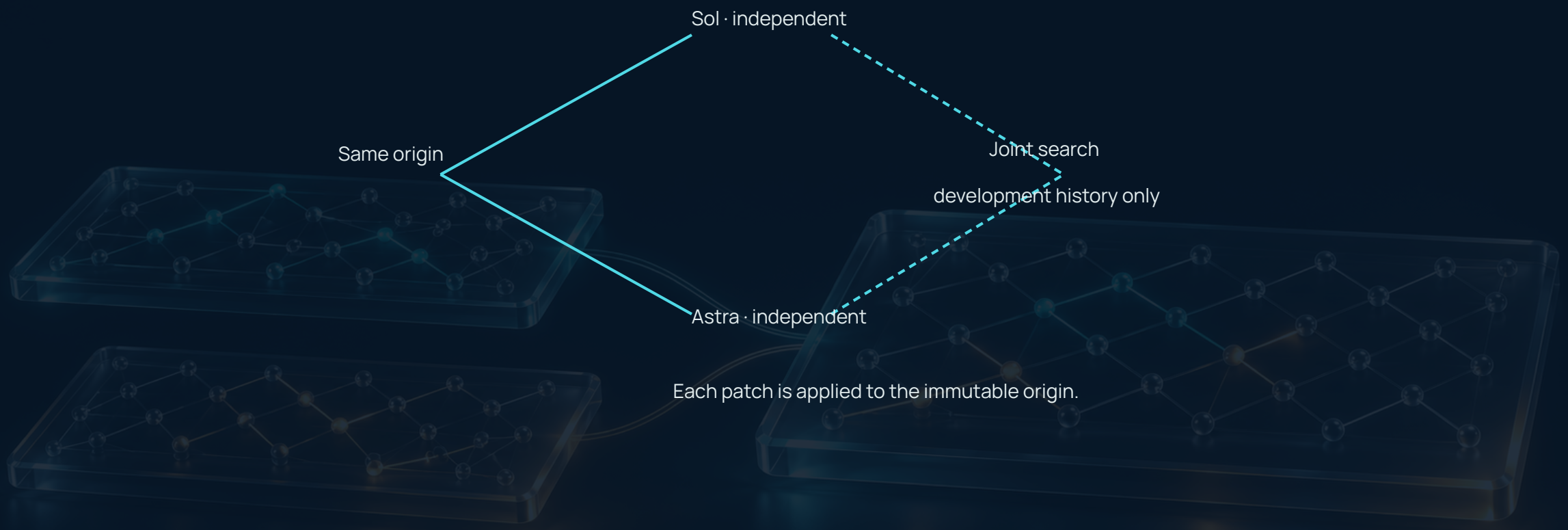
i: actual candidate index; di: measured DEV balanced accuracy (LM uses acc_norm).

Minimize the tuple from left to right: −di first means maximize the DEV score.

Ties prefer smaller L2 norm $\|\Delta i\|_2$, then fewer changed scalars #Δi, then earlier i. Candidate 0 is eligible.

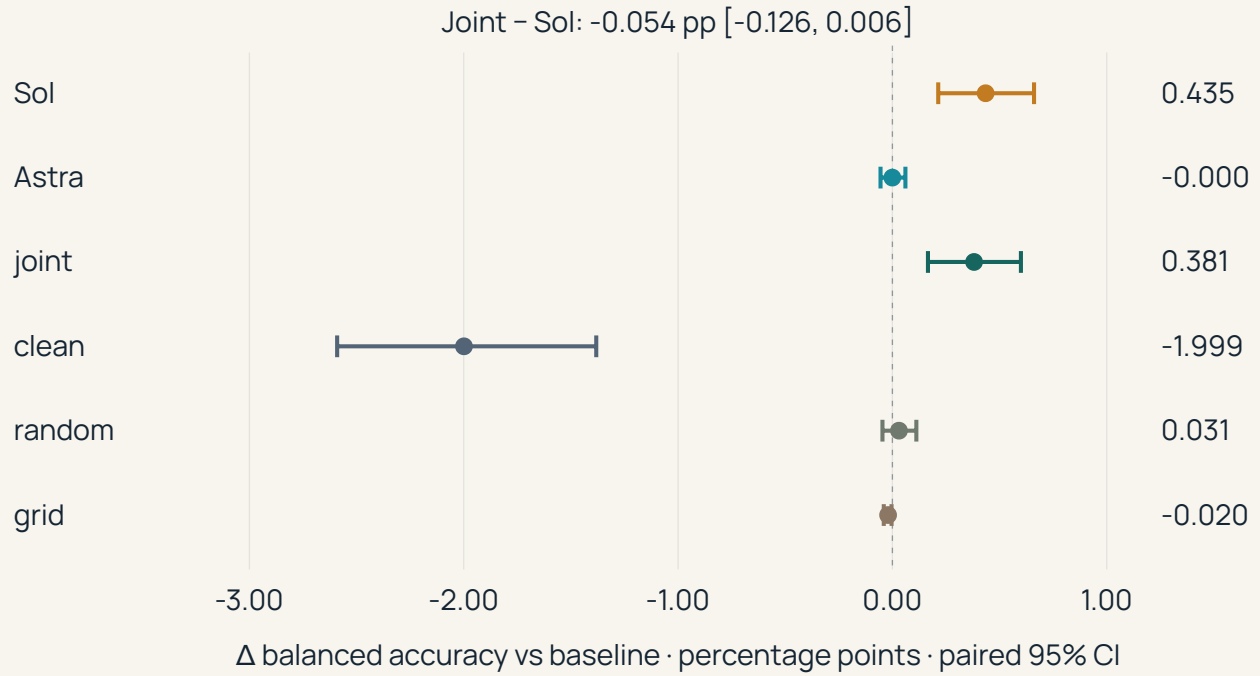
Share experience, keep the origin

The joint search receives development history, not a serial sum of solo edits.



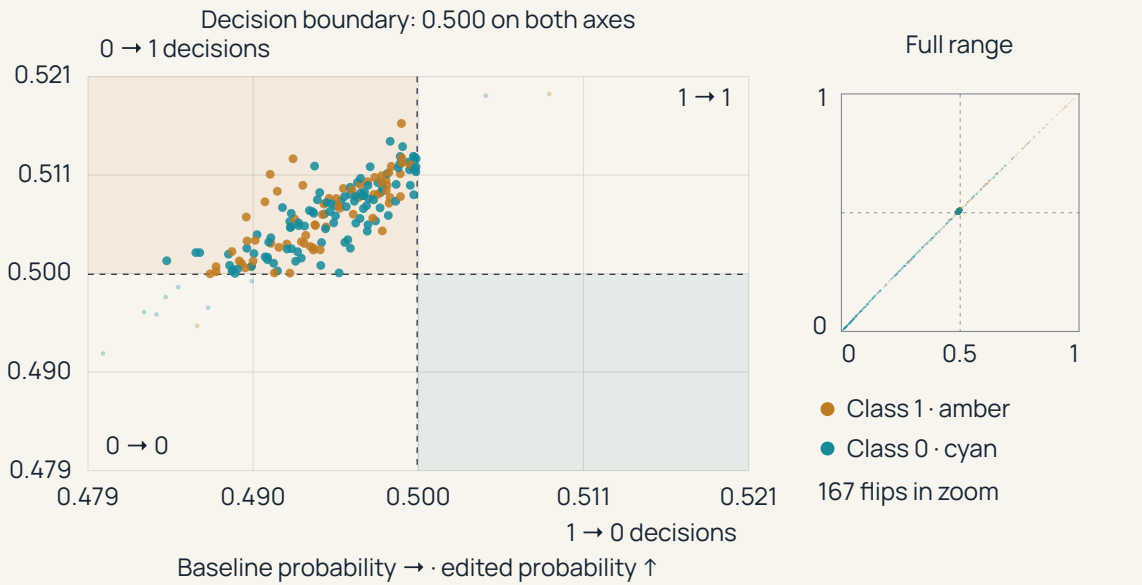
A small shift in balanced accuracy

Sol and joint move the primary metric upward; Astra is effectively flat.



Who changes sides?

Positive-class rescues come with negative-class regressions.



Class 1 · n=3,846

+65 rescues / **−0** regressions

Class 0 · n=12,435

+0 rescues / **−102** regressions

FINAL · Sol: 65 positive rescues / 0 regressions; 0 negative rescues / 102 regressions. BA gives each class half the weight; accuracy counts each row equally.

$$BA = \frac{1}{2}\left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP}\right)$$

TP / (TP+FN): positive recall; TN / (TN+FP): negative recall.

BA averages the two recalls equally, regardless of class size.

A rescue changes wrong → correct; a regression changes correct → wrong on the same saved row.

Better balance is not recovery

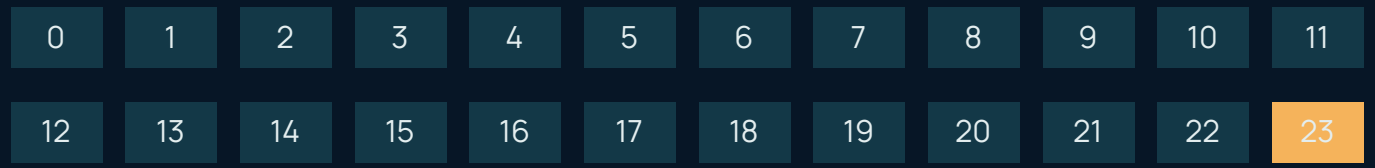
Clean has lower BA, but better accuracy, ranking and probability quality.

Adult final · 16,281 paired examples

State	BA ↑	Accuracy ↑	AUC ↑	Log-loss ↓	Brier ↓
baseline	78.586%	82.857%	0.889869	0.378285	0.120487
Sol	79.021%	82.630%	0.889981	0.382725	0.122237
Astra	78.586%	82.802%	0.889885	0.378581	0.120608
joint	78.967%	82.630%	0.889975	0.382383	0.122106
clean	76.587%	85.634%	0.907734	0.314733	0.100082
random	78.616%	82.808%	0.889981	0.37868	0.120655
grid	78.566%	82.827%	0.889862	0.376637	0.120089

From 153 nodes to 24 blocks

The architecture is schematic; intervention indices and measured operations are exact.

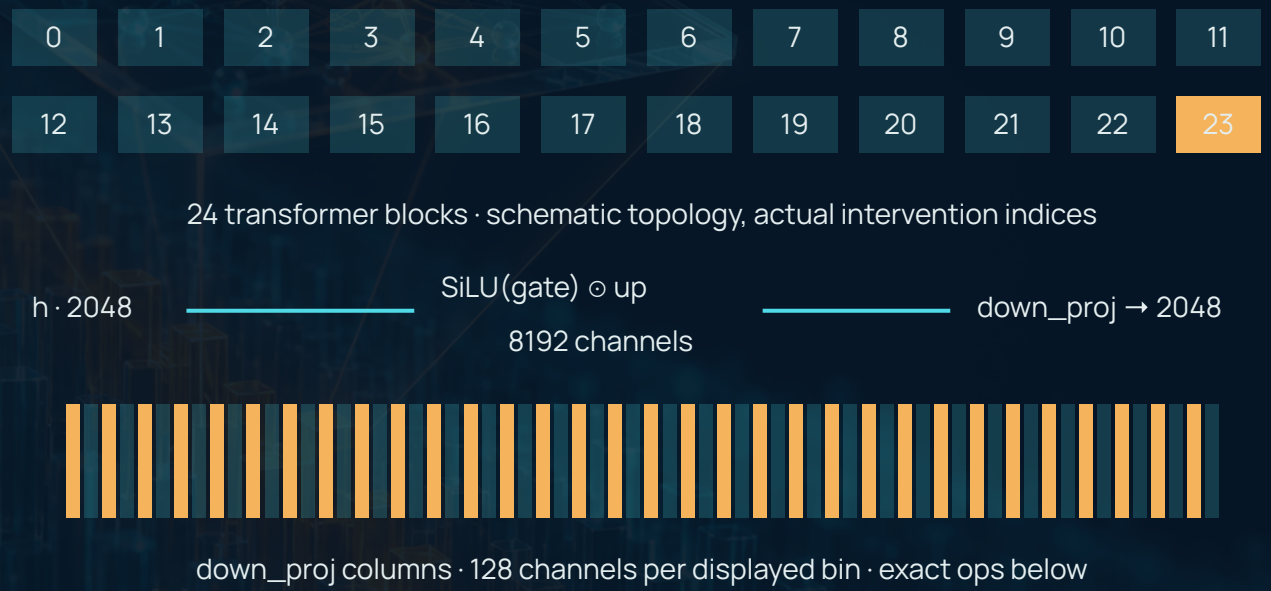


24 transformer blocks · schematic topology, actual intervention indices

Sol · candidate 1 · 65,536 changed scalars · L2 14.45690302 · DEV 75.000000%

Edit a channel contribution

2048 → 8192 → 2048, with selected columns scaled or sparse scalars changed.



Sol · candidate 1 · 65,536 changed scalars · L2 14.45690302 · DEV 75.000000%

$$u = \mathrm{SiLU}(W_g h) \odot (W_u h), \qquad y = W_d u$$

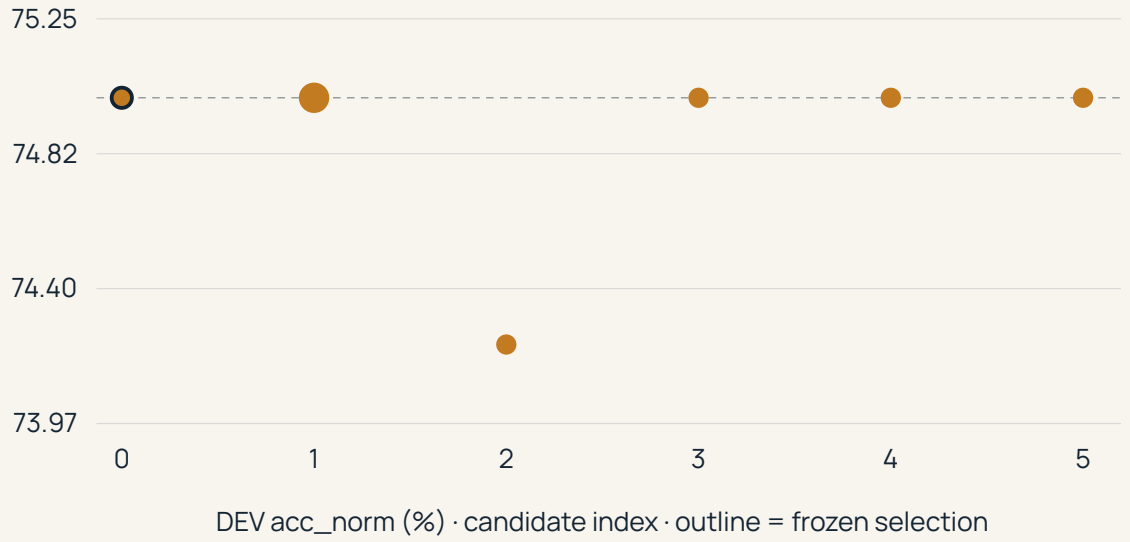
h: 2048 residual inputs; Wg and Wu map to 8192 intermediate channels.

SiLU(z) = z σ(z); ⊙ multiplies gate and up branches component by component.

u: 8192 gated channels. Wd (down_proj): 2048×8192; y: 2048 outputs.

A real search can select no change

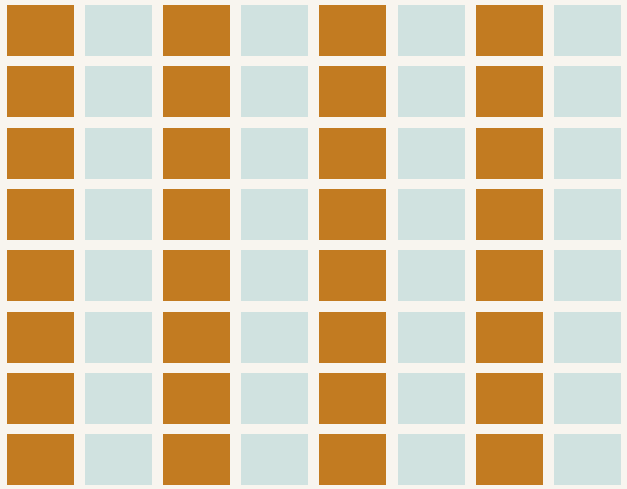
Nonempty candidates were measured. The unchanged origin still wins.



Block 23 · down_proj

65,536 changed scalars

L2 14.45690302



8192 channels · 128 / tile

Exact operations below

Sol · candidate 1 · 65,536 changed scalars · L2 14.45690302 · DEV 75.000000%

One state. Two benchmarks.

Four labels share one executed weight identity and one measured endpoint.

ARC-Easy

73.437500%

acc_norm · 376 / 512

Raw acc 77.343750%

396 / 512 correct

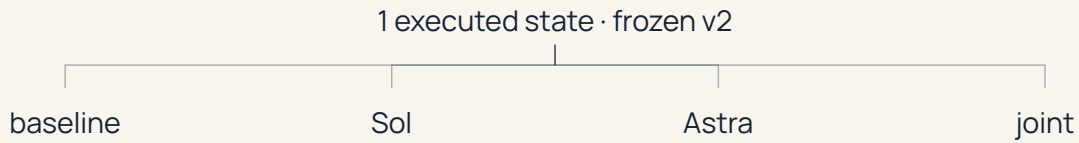
PIQA

78.515625%

acc_norm · 201 / 256

Raw acc 80.078125%

205 / 256 correct



$$s(c \mid q) = \frac{\sum_t \log p(c_t \mid q, c_{<t})}{\mathrm{len}_{chars}(c)}$$

q: question; c: choice text; ct: its continuation token at position t.

Sum log p(ct | q,c<t): continuation log likelihood; divide by character count, not token count.

Choose the first maximum normalized score; acc_norm is the fraction of correct choices. Raw accuracy omits the divisor.

Count work, not labels

Target forwards, editor calls and reused predictions are different quantities.

Recorded target work · native units kept separate

Phase	Examples	Forwards	Seconds	Editor calls
LM v2 DEV Sol	672	2,690	260.046	7
LM v2 DEV Astra	672	2,690	262.152	7
LM v2 DEV joint	416	1,666	161.753	14
LM v2 FINAL · unique state	768	2,558	248	0
Adult controls · CPU batches	48,843	3	0.005182	0

LM forwards are per-choice requests; Adult forwards are batched calls. One LM final state feeds four method labels; it is one observation, not four replications.

Intervention, decision, selection

Adult changes decisions. LM selection rejects change. Both are results.

01 A real weight intervention can change a decision.

02 Adult gains balanced accuracy, with a tradeoff.

03 SmolLM2 v2 selects the unchanged origin.

The figure is a view of the data

Saved numeric evidence rebuilds the metrics and visuals without a model call.

1. **Inspect** Frozen manifest and saved numeric predictions.
2. **Recompute** Native and offline reproduction guide.
3. **Rebuild** npm –prefix web run build
4. **Inspect** report, deck, figures and local source links.

Manifest 8f8869a113d93000bb8fffd547258b36ab955a6bf7a7f8ddbe589fbdbd231d38

DISCUSSION / Q&A

Small edits. Measured consequences.

Improving the chosen metric and choosing an edit are separate questions.

01 A real weight intervention can change a decision.

02 Adult gains balanced accuracy, with a tradeoff.

03 SmolLM2 v2 selects the unchanged origin.

Five minutes for questions · constraints / selection / interpretation

Technical appendix

From tensor to prediction

Adult forward pass

$$h_1 = \mathrm{ReLU}(W_1 x + b_1),\ h_2 = \mathrm{ReLU}(W_2 h_1 + b_2),\ p = \sigma(W_3 h_2 + b_3)$$

W_1:32×104; W_2:16×32; W_3:1×16. Biases match output dimensions.

Legal changes

$$W' = W_d + \Delta, \quad \Delta \odot M = 0, \quad \frac{\|W' - W\|_F}{\|W\|_F} \leq .05$$

M marks removed Adult entries. LM per-tensor relative Frobenius ≤ .05, global changed fraction ≤ .001; no overlapping operations.

LM application

$$W'_{:,j} = W_{:,j} + \mathrm{cast}_{dtype}(W_{:,j}(s-1))$$

s is the requested scale; sparse changes add cast(delta) to a scalar. Counts and norms use actual post-cast changes.

Choice selection

$$\hat{c}_{norm} = \arg\max_c \frac{\sum_t \log p(c_t \mid q, c_{<t})}{\mathrm{len}_{chars}(c)}$$

Raw acc omits the character divisor. Native first-argmax tie handling applies.

Metrics & paired uncertainty

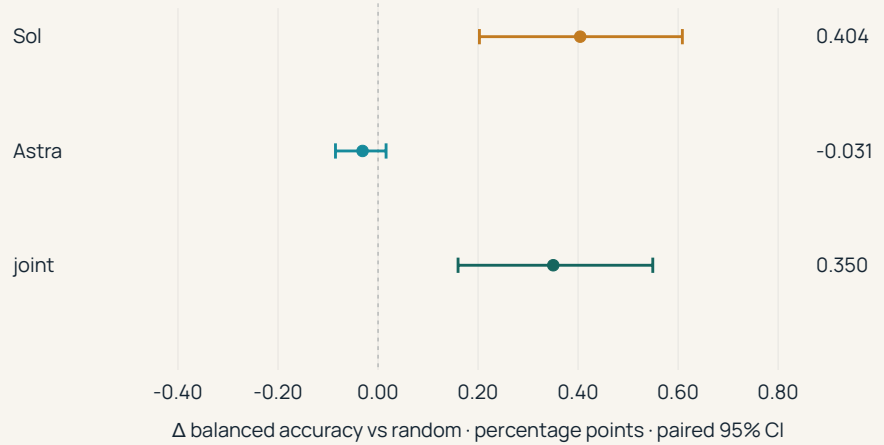
$$BA = \frac{1}{2}(\mathrm{TPR} + \mathrm{TNR}), \quad Acc = \frac{TP + TN}{N}$$

$$LL = -\frac{1}{N}\sum_i [y_i \log p_i + (1 - y_i)\log(1 - p_i)], \quad Brier = \frac{1}{N}\sum_i (p_i - y_i)^2$$

$$\delta_b = BA(y[I_b], p_A[I_b]) - BA(y[I_b], p_B[I_b])$$

The same row indices I_β are sampled in both conditions, with replacement separately within each Adult label class. Seed 101; 2000 draws; percentile 2.5/97.5%. Positive = income >50K (3,846); negative = ≤50K (12,435). ROC-AUC describes ranking; lower log-loss and Brier are better probability quality.

A rescue changes wrong to correct; a regression changes correct to wrong. Intervals concern paired examples in one session, not editor-session variation. The insufficient-damage flag is true; no BA recovery denominator is reported.



All final metrics

Adult final · 16,281 paired examples

State	BA ↑	Accuracy ↑	AUC ↑	Log-loss ↓	Brier ↓
baseline	78.586%	82.857%	0.889869	0.378285	0.120487
Sol	79.021%	82.630%	0.889981	0.382725	0.122237
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ARC-Easy

73.437500%

acc_norm · 376 / 512

Raw acc 77.343750%

396 / 512 correct

PIQA

78.515625%

acc_norm · 201 / 256

Raw acc 80.078125%

205 / 256 correct



All four LM labels share these v2 values and one weight identity. Paired acc/acc_norm differences vs baseline are zero, with 95% intervals [0,0] and zero rescues/regressions.

Methods, history & sources

One exploratory seed101. Adult solos start independently, ten allowed candidates each; joint has five attempts and sees only development history. All patches start at the same origin. Final feedback never selects or stops the search. Clean/random/grid supplements and LM v2 measurements are complete.

v1 LM prompt said float16 while execution used float32. v2 corrects that wording and uses fresh source sessions, but the analyst had seen v1 final results and the panels are reused. This is not an independent replication. Both historical v1 and main v2 selected the unchanged origin; scores remain separately versioned.

MPS float32 train8 was an engineering sensitivity check. No float32 train64 gate was run; this does not establish full-benchmark numerical equivalence. Exporting the snapshot made no model calls; original v2 measurement did complete.

Prior work: Rikiya Takehi, Jayden Teoh, Alex Zhang and Omar Khattab, Sparks of Weight-Space Programming, 2 November 2026. This independent study uses different targets and intervention domains.

Full methods · Versions and revisions · Reproduction · Source provenance

Exact prompts: v1 LM · v2 LM · Adult · Joint. Exact scorer and task continuation conventions: scorer reference.

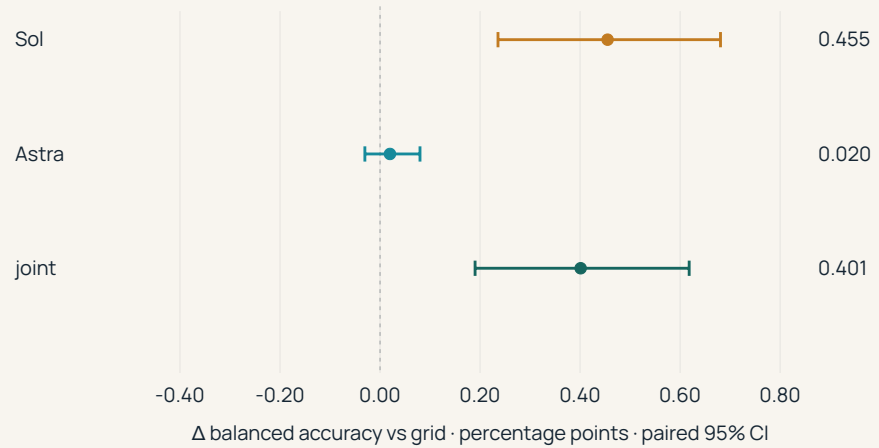
Manifest identity: 8f8869a113d93000bb8fffd547258b36ab955a6bf7a7f8ddbe589fbdbd231d38

Work & complete comparisons

Recorded target work · native units kept separate

Phase	Examples	Forwards	Seconds	Editor calls
LM v2 DEV Sol	672	2,690	260.046	7
LM v2 DEV Astra	672	2,690	262.152	7
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Adult controls · CPU batches	48,843	3	0.005182	0

LM forwards are per-choice requests; Adult forwards are batched calls. One LM final state feeds four method labels; it is one observation, not four replications.



Why balance and accuracy diverge

Let R_k and G_k count rescues and regressions in class k, and n_k be its fixed size. Recall changes by $(R_k - G_k)/n_k$. Substituting this into balanced accuracy gives:

$$\Delta BA = \frac{1}{2}\left(\frac{R_1 - G_1}{n_1} + \frac{R_0 - G_0}{n_0}\right), \quad \Delta Acc = \frac{R_1 + R_0 - G_1 - G_0}{n_1 + n_0}$$

For Sol, positive net correct = 65 / 3,846, negative net correct = -102 / 12,435. Thus ΔBA = 0.434901 percentage points while Δaccuracy = -0.227259 points. A negative-class regression costs less BA per example because that class is larger.

From a scalar to its local contribution

$$z_2' - z_2 = \Delta h_1, \qquad y' - y = (W'_{down} - W_{down})u$$

Adult h_1 is unchanged because fc1 is fixed. The preactivation difference enters ReLU, then the final linear layer and sigmoid. For the LM expression, u is the same local MLP input: the equation explains the edited projection, not all downstream logits. Exact post-cast differences, rather than requested decimals, define Δ. The intervention norm is the square root of the sum of squared executed scalar differences; the relative norm divides it by the origin tensor norm.